A Sub-THz Low-Cost Additive Manufactured Monolithic Geodesic H-Plane Horn Array Antenna

Mingzheng Chen, *Member, IEEE*, José Rico-Fernández, *Member, IEEE*, Hairu Wang, *Graduate Student Member, IEEE*, Cleofás Segura-Gómez, *Graduate Student Member, IEEE*, Francisco Mesa, *Fellow, IEEE*, and Oscar Quevedo-Teruel, *Fellow, IEEE*

***Abstract*—A monolithic geodesic H-plane horn array antenna that operates up to 170 GHz is achieved for the first time using a low-cost additive manufacturing (AM) technique. To reach high gain and symmetric beam, a truncated geodesic H-plane horn is used to obtain a narrow beam in the H-plane, while a 1:8 power divider built on parallel-plate waveguides is constructed to narrow the beam in the E-plane. A ray-tracing and physical-optics model is developed to facilitate the design, which is capable of computing the full radiation pattern, directivity, and gain (considering conductive losses) of geodesic H-plane horn array antennas with significant time efficiency and high degree of accuracy. The adopted metal-only laser powder–bed fusion AM technique is especially suitable for fast prototyping structures with intricate shapes at a low cost. However, special adaptations are still considered in the design to ensure a successful fabrication of the prototype operating at the *D*-band. The prototype maintains good frequency stability from 110 to 170 GHz with a return loss better than 10 dB and a symmetric pencil beam. The measured data show a maximum realized gain of 29 dBi, a maximum aperture efficiency of 67% (calculated using *realized gain*), and a maximum radiation efficiency of 86%.**

***Index Terms*—Additive manufacturing (AM), array antenna, *D*-band, geodesic H-plane horn, high gain, laser powder–bed fusion (LPBF), physical optics, ray tracing, symmetric beam.**

I. Introduction

TERAHERTZ (THz) and sub-THz frequencies are demanding increasing research interest due to the potentials offered by the immense available bandwidths. Possible applications include but are not limited to high-resolution imaging, hyperaccurate location and detection, and high-speed wireless communication systems [1], [2], [3]. At such high frequencies, the properties of high gain, high efficiency, and broad operating bandwidth are generally desired for the adopted antennas. In particular, high-gain antennas are essential to compensate for increased path loss in the THz and sub-THz bands, which helps to enhance the signal-to-noise ratio in radar and communication systems [4]. Moreover, a narrow and symmetric beam and a broad bandwidth are vital to provide high spatial and range resolutions for high-frequency radar and imaging systems [5].

The design of high-performance antennas plays a crucial role in advancing sub-THz systems. Numerous antenna configurations have been reported for related applications in recent literature, such as antennas based on printed circuit boards (PCBs) [6], [7], silicon substrates [8], [9], and metallic waveguides [10], [11], among others. Antennas constructed using PCBs, including those utilizing microstrip, substrate integrated waveguide (SIW), and low-temperature co-fired ceramic (LTCC) technologies, are characterized by their lightweight nature and low-profile design. Nevertheless, their principal drawbacks include substantial dielectric losses, limited power handling capacity, and considerable signal leakage at sub-THz frequencies. Consequently, achieving radiation efficiencies greater than 50% remains challenging. Solutions that employ silicon-based and metallic waveguides, commonly produced through microelectromechanical systems (MEMSs) micromachining and computer numerical control (CNC) machining, often exhibit high efficiency. Nonetheless, constructing intricate prototypes with precision necessitates machining techniques that can be both time-intensive and expensive.

An alternative solution at sub-THz is quasioptical structures, including lenses and reflectors [4], [12], [13], [14], [15]. These quasioptical devices are designed based on optical principles and concepts. Therefore, by eliminating the complex beamforming network (BFN), quasioptical antennas are especially suitable for high-frequency applications, because of their generally simple designs and high tolerance to manufacturing errors [16]. A promising quasioptical approach applicable in sub-THz is the utilization of geodesic lenses and horns [17], [18], [19], [20]. Geodesic structures make use of the out-of-plane profile in a parallel-plate waveguide (PPW) to tailor physical path length differences that control the propagation of electromagnetic

Received 31 July 2025; revised 11 September 2025 and 13 October 2025; accepted 16 October 2025. Date of publication 20 October 2025; date of current version 23 February 2026. The work of Oscar Quevedo-Teruel was supported by the VR Project for the manufacture and measurement of the prototype under the call "Research project grant within natural and engineering sciences" under Grant 2022-03865. The work of Francisco Mesa was supported in part by the MICIU/AEI/10.13039/501100011033 and ERDF/EU under Grant PID2023-148281NB-I00 and in part by the Spanish Ministerio de Ciencia, Innovación y Universidades under Grant Programa Movilidad Senior PRX23/00102. *(Corresponding author: Mingzheng Chen.)*

Mingzheng Chen, Hairu Wang, and Oscar Quevedo-Teruel are with the Division of Electromagnetic Engineering and Fusion Science, KTH Royal, Institute of Technology, SE-100 44 Stockholm, Sweden (e-mail: mzchen@kth.se; hairu@kth.se; oscarqt@kth.se).

José Rico-Fernández is with Northern Waves AB, SE-115 43 Stockholm, Sweden (e-mail: pepe.rico@northern-waves.com).

Cleofás Segura-Gómez is with the Department of Signal Theory, Telematics and Communications. Research Centre for Information and Communication Technologies, University of Granada 18071 Granada, Spain (e-mail: cleofas@ugr.es).

Francisco Mesa is with the Department of Applied Physics 1, ETS Ingeniería Informática, Universidad de Sevilla, 41012 Seville, Spain (e-mail: mesa@us.es).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TTHZ.2025.3623926.

Digital Object Identifier 10.1109/TTHZ.2025.3623926

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

waves along the geodesic surface [21], [22]. Hence, geodesic structures are particularly attractive for high-frequency applications because of their simple and fully metallic configuration. In addition, regarding production, geodesic lenses and horns are readily manufacturable in a monolithic form through the use of metal-only laser powder–bed fusion additive manufacturing (LPBF-AM). In the literature, a family of LPBF-AM geodesic designs up to 110 GHz has been reported [23], [24], [25], [26], [27]. However, these PPW-based designs can focus the beam only in the H-plane, resulting in a fan-shaped beam and a maximum gain that hardly exceeds 25 dBi. In [28] and [29], four geodesic lenses are vertically stacked to form a linear array to narrow the beamwidth in the E-plane. However, the resulting beams are asymmetric with different beamwidths in the H- and E-planes, and the designs are relatively bulky.

The literature discusses two effective horn designs for achieving a symmetric beam: horn array antennas [10], [11], [30] and diagonal horn antennas [31], [32], [33], [34]. Scaling up high-gain horn array antennas that rely on extensive waveguide feed networks to sub-THz frequencies is generally challenging and costly. Conversely, diagonal horns, due to their remarkably straightforward design, have been commercially accessible at frequencies exceeding 1 THz. However, these horns exhibit two main drawbacks. First, attaining high gain and high aperture efficiency with a short axial length in diagonal horns is challenging due to unavoidable phase errors introduced as the flare opens. Second, their inherently low polarization purity results in up to 10% of radiated power being cross-polarized [31], [32]. This problem might be alleviated through the optimal configuration of the horn profile [33].

Here, we present a low-cost monolithic geodesic H-plane horn array antenna with high gain and a symmetric beam in the *D*-band manufactured using LPBF-AM. Instead of stacking the horns, the array in the E-plane is achieved by connecting the single geodesic horn to a 1:8 PPW power divider. Taking into account the large electric size of the structure, the design of the geodesic horn array is elusive only with commercial full-wave simulators. Therefore, we extend the ray-tracing and physical-optics (RT-PO) model reported in [18] for a single geodesic H-plane horn to design geodesic H-plane horn array antennas with a symmetric beam. The experimental results of the prototype align well with the full-wave simulations and the numerical evaluations. Most importantly, the design achieves high gain, symmetric beam shape, high efficiency, high polarization purity, and a broad operating bandwidth, which are the merits that make it highly suitable in the sub-THz frequencies. This is the first time that the LPBF-AM technique has been successfully applied to a geodesic design up to 170 GHz, which is the outcome of both the advancement of the manufacturing technique in accuracy and the robustness of geodesic designs to manufacturing tolerances. Unlike conventional machining, which becomes very expensive and time consuming at these frequencies, LPBF-AM provides the required precision and geometric flexibility while drastically reducing both cost and lead time. We have shown that the LPBF-AM technique has the potential for fast and low-cost prototyping in the sub-THz bands.

The rest of this article is organized as follows. In Section II, we describe the RT-PO model to analyze geodesic H-plane horn array antennas. In Section III, the RT-PO model is used to design a geodesic H-plane horn array antenna with a symmetric beam in the *D*-band. Next, the full-wave and experimental results of the LPBF-AM prototype are reported in Section IV. Finally, Section V concludes this article.

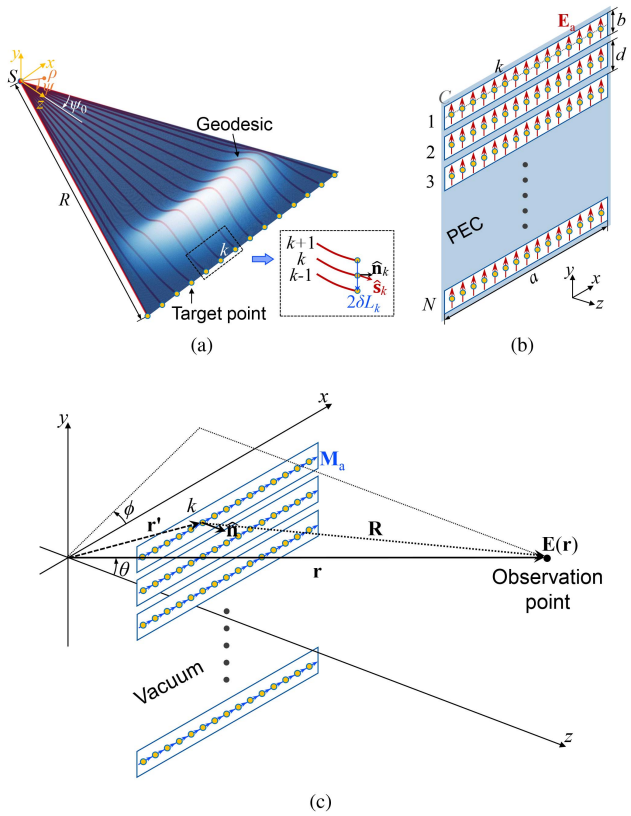


Fig. 1. (a) Geodesics in a triangular curved surface traced from the source point S to target points in the aperture. (b) N number of equally distributed apertures of a geodesic H-plane horn array. (c) Distribution of the equivalent magnetic currents and geometric parameters for numerical evaluation of far-field radiation characteristics.

II. RT-PO Model

A. Geodesic H-Plane Horn

A geodesic H-plane horn can be defined by an isosceles triangular curved surface that is used to tailor the path lengths of rays (or geodesics) from the source S to the target points in the aperture, as illustrated in Fig. 1(a). The triangle has a side length R and a half-flare angle ψ_0. A well-designed geodesic surface collimates the rays as they arrive parallel at the aperture, especially in the middle part, where the field power concentrates. According to the definition of the coordinate system in Fig. 1(a), an appropriate out-of-plane height y-profile to describe such a surface is [18]

$$y(\rho,\psi) = A_{\mathrm{h}} R \exp\left[-\left(\frac{\rho-\rho_0}{\eta}\right)^2\right] \times \left[\left(\frac{\cos\left(\frac{\pi\psi}{\psi_0}\right)+1+\delta}{2}\right)^p - \left(\frac{\delta}{2}\right)^p\right] \quad (1)$$

in which $A_{\mathrm{h}}, \rho_0, \eta$, and p are four design parameters and $\delta = 0.01$ is a constant. The profile has a Gaussian distribution along the ρ-direction and a quasicosine distribution along the ψ-direction. Consequently, rays situated in the central region of the geodesic surface travel an additional path length in the vertical y-direction relative to the rays on the periphery. This approach makes it possible to equal the rays' phase at the aperture.

B. RT-PO Model for Geodesic H-Plane Horn Array

Our initial task is to obtain the electric field in the aperture of a single geodesic H-plane horn. Assuming that the field propagates as a TEM mode in the present PPW environment, this task can be divided into the following three steps [18], [35].

1) Find the ray trajectory from the source S to any target point k in the aperture along the geodesic surface using the specialized Python library `gdist` [36].
2) Evaluate the phase ϕ_k of the electric field at the kth target point using the path length ξ_k of the traced ray. This phase is given by $\phi_k = -k_0\xi_k$ where k_0 is the wavenumber in the free space.
3) Obtain the amplitude A_k of the electric field at the kth target point applying the ray-tube power conservation theory.

After these operations, the TEM electric field at the kth target point in the horn aperture is written as

$$\mathbf{E}_k = A_k \mathrm{e}^{-\mathrm{j}k_0\xi_k}\hat{\mathbf{y}}\,. \tag{2}$$

Next, the single horn is transformed into an array through a 1:N PPW power divider in the E-plane. Fig. 1(b) shows the aperture of the resulting array, which consists of N subapertures with the size of $a \times b$ and distributed equally in the E-plane with an interval of d. For operational purposes, the array aperture is assumed to be situated on a screen made of a perfect electric conductor. If the power divider is ideal, that is, the divided fields have an equal amplitude and phase, the electric field distributions in all subapertures are the same and can be denoted as $\mathbf{E}_{\mathrm{a}}$ (the discretized $\mathbf{E}_{\mathrm{a}}$ at the kth target point is $\mathbf{E}_k$). As the horn aperture is located in the xy-plane, the aperture electric field can be written as

$$\mathbf{E}_{\mathrm{a}}(\mathbf{r}') = E_{\mathrm{a}}(x')\hat{\mathbf{y}} \tag{3}$$

which varies only in the x-direction and is constant in the y-direction.

Finally, we apply the Schelkunoff equivalence principle [37] and the image principle [38] to the horn aperture region on the basis of the PO approximation. The actual source $\mathbf{E}_{\mathrm{a}}(\mathbf{r}')$ is then replaced by the following equivalent magnetic source located in vacuum:

$$\mathbf{M}_{\mathrm{a}}(\mathbf{r}') = -2\hat{\mathbf{n}} \times \mathbf{E}_{\mathrm{a}}(\mathbf{r}') = 2E_{\mathrm{a}}(x')\hat{\mathbf{x}} \tag{4}$$

where $\hat{\mathbf{n}}$ is the unit vector normal to the aperture ($\hat{\mathbf{n}} = \hat{\mathbf{z}}$), as illustrated in Fig. 1(c). The subapertures are symmetrically arranged with respect to the z-axis. As detailed in [39], the electric far-field $\mathbf{E}(\mathbf{r})$ at the observation point is given by [38], [40]

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}k_0 \frac{\mathrm{e}^{-\mathrm{j}k_0 r}}{4\pi r}\hat{\mathbf{r}} \times \left[-\widetilde{\mathbf{L}}\right] \tag{5}$$

where $\widetilde{\mathbf{L}}$ stands for (considering N subapertures)

$$\widetilde{\mathbf{L}}(\theta,\phi) = \widetilde{L}_\theta\hat{\boldsymbol{\theta}} + \widetilde{L}_\phi\hat{\boldsymbol{\phi}} = \sum_{n=1}^{N}\int_{S'_n} \mathbf{M}_{\mathrm{a}}(\mathbf{r}'_n)\,\mathrm{e}^{\mathrm{j}\mathbf{k}_0\cdot\mathbf{r}'_n}\mathrm{d}S' \tag{6}$$

with $\mathbf{k}_0 = k_0\hat{\mathbf{r}} = k_x\hat{\mathbf{x}} + k_y\hat{\mathbf{y}} + k_z\hat{\mathbf{z}}$ being a wavevector with components $k_x = k_0\cos\phi\sin\theta$, $k_y = k_0\sin\phi\sin\theta$, and $k_z = k_0\cos\theta$. Combining (5) and (6), we can write the far-region electric field in spherical components as

$$E_r = 0 \tag{7}$$

$$E_\theta = -\mathrm{j}k_0\frac{\mathrm{e}^{-\mathrm{j}k_0 r}}{4\pi r}\widetilde{L}_\phi \tag{8}$$

$$E_\phi = \mathrm{j}k_0\frac{\mathrm{e}^{-\mathrm{j}k_0 r}}{4\pi r}\widetilde{L}_\theta\,. \tag{9}$$

Introducing (4) into (6) and solving the integration, we obtain the following expressions for electric far-field components:

$$E_\theta = \mathrm{j}k_0\sin\phi\frac{\mathrm{e}^{-\mathrm{j}k_0 r}}{2\pi r}\frac{b\sin Y}{Y}\,\mathrm{AF}(\theta,\phi)\,\widetilde{E}_{\mathrm{a}}(\theta,\phi) \tag{10}$$

$$E_\phi = \mathrm{j}k_0\cos\theta\cos\phi\frac{\mathrm{e}^{-\mathrm{j}k_0 r}}{2\pi r}\frac{b\sin Y}{Y}\,\mathrm{AF}(\theta,\phi)\,\widetilde{E}_{\mathrm{a}}(\theta,\phi) \tag{11}$$

where

$$Y = k_0(b/2)\sin\theta\sin\phi \tag{12}$$

$$\mathrm{AF}(\theta,\phi) = \sum_{n=1}^{N}\mathrm{e}^{\mathrm{j}k_0\sin\theta\sin\phi\left(\frac{N+1}{2}-n\right)d} \tag{13}$$

$$\widetilde{E}_{\mathrm{a}}(\theta,\phi) = \int_{-a/2}^{a/2} E_{\mathrm{a}}(x')\mathrm{e}^{\mathrm{j}k_0 x'\sin\theta\cos\phi}\mathrm{d}x' \tag{14}$$

with $\mathrm{AF}(\theta,\phi)$ being the array factor in the E-plane of the horn array. The integral $\widetilde{E}_{\mathrm{a}}(\theta,\phi)$ can be numerically calculated as the following sum along the middle straight line C of the subaperture using the electric field at the target points given in (2):

$$\widetilde{E}_{\mathrm{a}}(\theta,\phi) \approx \sum_k A_k \mathrm{e}^{-\mathrm{j}k_0(\xi_k - x_k\sin\theta\cos\phi)}\delta L_k \tag{15}$$

where $\delta L_k = (x_{k+1} - x_{k-1})/2$ is the width of the kth ray tube in the aperture, as shown in Fig. 1(a). A simple Simpson rule with integration intervals of $\delta L_k \lesssim \lambda/10$ has been tested to provide good convergence and considerably accurate results [41].

Moreover, similar to [39, Eq. (31)] and taking into account the presence of N subapertures, the antenna directivity of the horn array is given by

$$D(\theta,\phi) \approx \frac{4\pi r^2\left(|E_\theta|^2 + |E_\phi|^2\right)}{Nb\sum_k A_k^2(\hat{\mathbf{s}}_k\cdot\hat{\mathbf{n}})\delta L_k} \tag{16}$$

with $\hat{\mathbf{s}}_k$ being the unit vector along the direction of the kth ray at the horn aperture. The factor $(\hat{\mathbf{s}}_k\cdot\hat{\mathbf{n}})$ accounts for the oblique incidence of the rays in the horn aperture.

Finally, the antenna gain considering the conductive losses can also be evaluated using the RT-PO model. To achieve this goal, we simply replace A_k in (15) with

$$A_{k_\text{loss}} = A_k \mathrm{e}^{-\alpha_\text{G}\xi_k} \mathrm{e}^{-\alpha_\text{D}\xi_\text{D}} \tag{17}$$

in which α_G and α_D are the attenuation coefficients of the geodesic horn and the power divider, respectively, and ξ_D is the path length inside the power divider. Since both the geodesic horn and the power divider are based on PPW, their attenuation coefficients are given by [24], [42]

$$\alpha = \frac{R_\text{s}}{Z_0 h_\text{PPW}} \tag{18}$$

where R_s is the surface resistance of the conductor, Z_0 is the characteristic impedance of the PPW ($\approx 377\,\Omega$), and h_PPW is the height of the PPW air channel. R_s of the PPW is further given as

$$R_\text{s} = \sqrt{\frac{\omega\mu_0}{2\sigma}} \tag{19}$$

with ω being the angular frequency, μ_0 the permeability in vacuum, and σ the conductivity of the building material. Therefore, considering conductive losses in (10) and (11), the antenna gain is calculated as

$$G(\theta,\phi) \approx \frac{4\pi r^2 \left(|E_{\theta_\text{loss}}|^2 + |E_{\phi_\text{loss}}|^2\right)}{Nb\sum_k A_k^2(\hat{\mathbf{s}}_k \cdot \hat{\mathbf{n}})\delta L_k}\,. \tag{20}$$

Certainly, we can obtain the conduction efficiency as [38]

$$e_\text{c} = \frac{\max\{G(\theta,\phi)\}}{\max\{D(\theta,\phi)\}}\,. \tag{21}$$

It should be noted that the approximate approach proposed for the geodesic H-plane horn array, which combines the RT-PO model for a single H-plane horn and the E-plane array factor, can be easily extended to more general PPW-based arrays.

III. Geodesic H-Plane Horn Array Design

In this section, the design of geodesic H-plane horn arrays with a symmetric pencil beam is elaborated. In particular, the design target is to achieve the same beamwidth in the E- and H-planes. First, the basic dimensions of the geodesic H-plane horn array are determined using the proposed in-house RT-PO tool. Subsequently, all components, such as the feed transition, geodesic horn, power divider, and flare, are designed taking into account the manufacturing tolerances associated with the LPBF-AM process in the *D*-band.

A. Symmetric Beam Design

The symmetric beam design process utilizing the RT-PO model can be divided into the following three steps.

1) For a given frequency range of the design $[f_\text{min}, f_\text{max}]$, decide the height b, the separation d, and the number N of subapertures.
2) Calculate the E-plane radiation pattern at the central operating frequency f_0 and obtain $\phi_{3\,\text{dB}}$, the 3 dB beamwidth.
3) Select R and ψ_0 for an H-plane horn and design the geodesic profile according to (1) to achieve the same $\phi_{3\,\text{dB}}$ in the H-plane.

In the first step, the selection of b and d is restricted by three main factors as follows.

a) b has to be large enough ($\approx \lambda$) to ensure a good impedance match with the free space.
b) d needs to be small to avoid severe grating lobes (ideally $\leq \lambda$), especially at f_max.
c) The width of the metal wall must be greater than the manufacturing limit ($d - h \geq W_\text{min}$).

The minimum wall width is $W_\text{min} = 0.2$ mm in the applied LPBF-AM technique. In this work, the target design frequency range is 110 – 170 GHz (covering the entire *D*-band) with $f_0 =$ 140 GHz. Therefore, we choose $b = 1.71$ mm and $d = 1.93$ mm, which correspond to $b = 0.8\lambda_0$ and $d = 0.9\lambda_0$ at f_0 and give rise to a wall width of 0.22 mm. The number of subapertures is selected to be $N = 8$.

To determine the electric far field in the E-plane of the horn array under the selected parameters, we take $\phi = \pi/2$ in (10) and (11), resulting in $\mathbf{E} = E_\theta(r,\theta)\hat{\boldsymbol{\theta}}$ with

$$E_\theta(r,\theta) = \mathrm{j}k_0 b \frac{\mathrm{e}^{-\mathrm{j}k_0 r}}{2\pi r} \frac{\sin\left(k_0(b/2)\sin\theta\right)}{k_0(b/2)\sin\theta} \times \sum_{n=1}^{N} \mathrm{e}^{\mathrm{j}k_0 \sin\theta((N+1)/2-n)d} \sum_k A_k \mathrm{e}^{-\mathrm{j}k_0\xi_k}\delta L_k\,. \tag{22}$$

The resulting E-plane directivity pattern at $f_0 = 140$ GHz is presented in Fig. 2 in a solid blue curve; the $\phi_{3\,\text{dB}}$ achieved is 6.9°. The first side lobe is approximately 13 dB lower than the main beam, which is a typical behavior of an in-phase equal-amplitude array. Note that the side-lobe level could be reduced by applying amplitude tapering, a study that is beyond the scope of the present work.

Finally, to obtain the same $\phi_{3\,\text{dB}} = 6.9°$ in the H-plane, the horn geometry is chosen to be $R = 32.14$ mm ($15\lambda_0$) and $\psi_0 = 24°$ after some iterations. The parameters of the geodesic profile (1) designed are $A_\text{h} = 0.16$, $\rho_0 = 0.78$, $\eta = 0.072$, and $p = 0.138$. Applying $\phi = 0$ to (10) and (11), we find that the

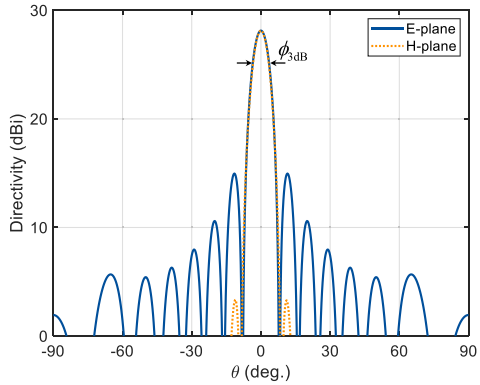


Fig. 2. Directivity patterns of the geodesic H-plane array in the E- and H-planes at 140 GHz obtained using the RT-PO model.

electric far field in the H-plane is given by $\mathbf{E} = E_\phi(r,\theta)\hat{\phi}$ with

$$E_\phi(r,\theta) = \mathrm{j}k_0 N b \frac{\mathrm{e}^{-\mathrm{j}k_0 r}}{2\pi r} \times \sum_k A_k \mathrm{e}^{-\mathrm{j}k_0(\xi_k - x_k \sin\theta)} \delta L_k \,. \quad (23)$$

In our implemented Python code running on a computer with an Intel i9-10900K CPU, the computation time required to calculate the H-plane radiation pattern of a specific geodesic shape is less than 5 s. In the computation, no GPU or parallelization was used and the usage of RAM was 3 GB. For each horn with given R and ψ_0, it takes some trials to find the appropriate geodesic profile with a directive beam. Therefore, with a preliminary estimate of the aperture width ($2R\sin\psi_0$) that is potentially needed and a few tests, it takes less than 1 h to obtain the final dimensions of the geodesic horn using the RT-PO tool. The yellow dotted curve in Fig. 2 is the calculated directivity pattern of the designed horn array in the H-plane. We can observe that the main beam overlaps almost perfectly with that in the E-plane, resulting in a symmetric pencil beam. Moreover, due to the inherent tapering amplitude, as discussed in [27], the level of the side lobe is -25 dB in the H-plane. Finally, the calculated peak directivity value of the horn array is 28.1 dBi at 140 GHz. The calculated antenna gain is 27.7 dBi, which corresponds to a conduction efficiency of -0.4 dB, considering $\sigma = 1.68 \times 10^7$ S/m for the AlSi10Mg aluminum alloy used to build the antenna [43].

B. Antenna Design

An illustration of the monolithic geodesic H-plane horn array antenna specially tailored for the LPBF-AM technique is shown in Fig. 3(a). To reduce the manufacturing difficulty at such high frequencies, the height h of the air channel following the geodesic shape is selected to be 0.6 mm, which corresponds to 0.34λ at the highest operating frequency of 170 GHz. This ensures the monomodal propagation of TEM waves within the horn given that $h < \lambda/2$ is satisfied. On the feed side, a standard WR-6 flange with a cross-sectional dimension of 1.65×0.83 mm^2 is used for the measurements. Therefore, a smooth sinusoidal feed transition with a total length of 4.5 mm is designed to adjust the height from $h_0 = 0.83$ mm to $h = 0.6$ mm, as shown in the inset of Fig. 4(a). In this figure, it also shows that the reflection coefficient of the transition is below -35 dB in the entire D-band. Similarly, we design a sinusoidal flare to transform the height of the air channel from $h = 0.6$ mm to $b = 1.71$ mm with a total length of 4.5 mm on the aperture side. As it can be observed in Fig. 4(a), the reflection coefficient of the flare is better than -20 dB in most of the D-band.

Next, a 1:8 PPW power divider is designed to connect the geodesic H-plane horn with eight flares on the aperture side. As shown in Fig. 3(a), the open angle of each branch in the power divider is $\alpha = 28.4°$, which ensures that the model can be vertically printed along the z-axis with the flange touching the printing bed ($\alpha < 45°$). The height of the air channel in each branch gradually increases from 0.3 to 0.6 mm, ensuring a good impedance match. The S parameters of the power divider are shown in Fig. 4(b). We can observe that the reflection coefficient

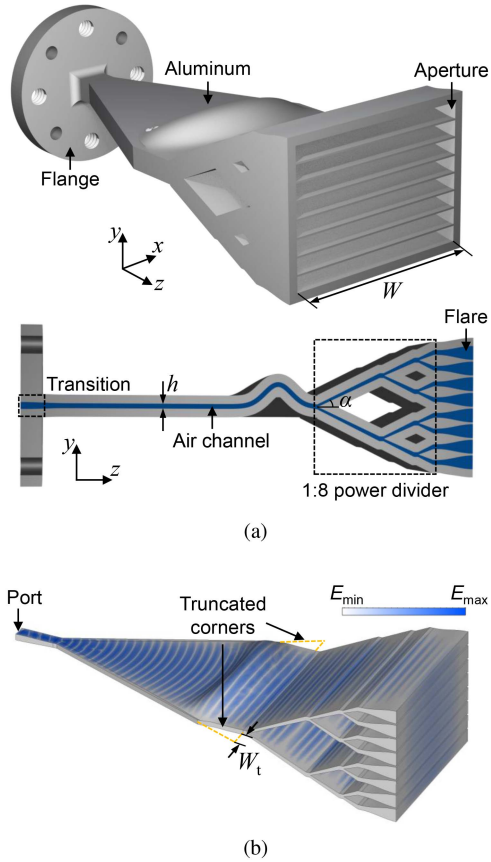


Fig. 3. (a) Illustration of the geodesic H-plane horn array antenna design and the cross-sectional view with the air channel, including the feed transition, the 1 : 8 power divider, and the flares indicated. (b) Electric field distribution on the surface of the air channel, with truncated corners indicated.

$|S_{11}|$ is lower than -15 dB in most of the operating band. Since port i is symmetric with port $(11-i)[i = 2,\ldots,9]$, only half of the transmission coefficients $|S_{i1}|$ are displayed in Fig. 4(b). It can be seen that all of these $|S_{i1}|$ are around -9 dB with small variations, showing an even division of transmitted power.

As discussed in [27], the two triangular corners of the geodesic horn can be truncated without degrading its performance due to the very low amplitude of the electric field in these areas. Consequently, a truncated width of $W_t = 2.5$ mm is selected as shown in Fig. 3(b), which represents a reduction of 19.1% in the aperture width, resulting in a final aperture width of $W = 21.15$ mm. Moreover, we can also observe from Fig. 3(b) that the cylindrical wave from the waveguide port is gradually transformed into a nearly planar wave after the geodesic shape and well maintained until the aperture.

Finally, a conformal aluminum block is constructed surrounding the air parts with a thickness of 1 mm to ensure mechanical robustness. The final model was analyzed in *CST Microwave Studio*. Taking into account the large electric size of the structure, CST took approximately 6 h to complete the simulation. Fig. 5 presents the radiation patterns of the horn array simulated using CST and evaluated by the proposed RT-PO model. The patterns show excellent agreement in the shape of the main beam and the side lobes in the H- and E-planes. In addition, the antenna directivity and gain versus frequency of the final design obtained

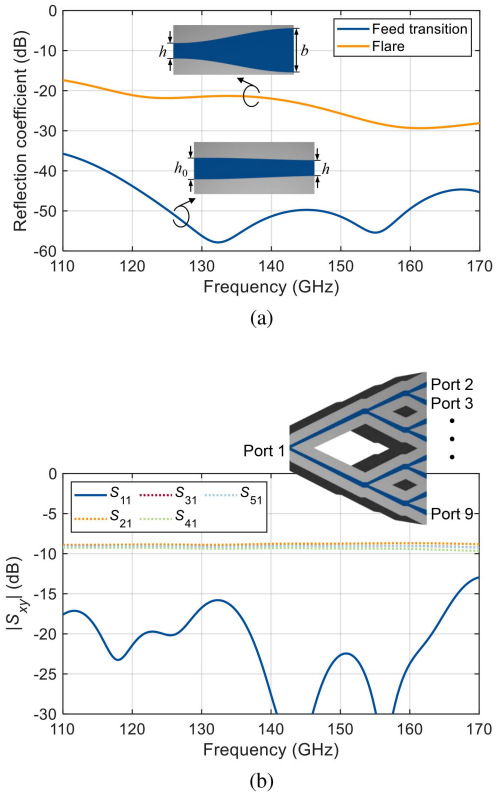


Fig. 4. (a) Simulated reflection coefficients of the feed transition and the flare. (b) Simulated reflection and transmission coefficients of the $1:8$ PPW power divider.

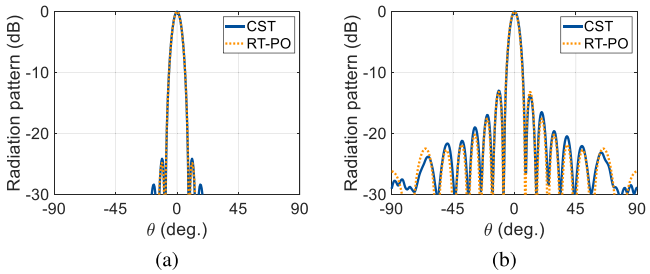


Fig. 5. Comparison of the radiation patterns at 140 GHz obtained using CST and RT-PO in the (a) H-plane and (b) E-plane.

using CST and the RT-PO model are shown in Fig. 6. We can see that the calculated RT-PO directivity values are close to the simulated ones with a small difference of $0.2-0.5$ dB, which is good enough for preliminary approximations. Similar agreement in antenna gain can also be observed assuming the same conductivity of $\sigma = 1.68 \times 10^7$ S/m in both the CST and RT-PO models, giving rise to an average conduction efficiency of -0.6 and -0.5 dB, respectively. It is noteworthy that, in contrast to the extensive hours required by the CST simulation, our RT-PO model takes only seconds to analyze the final structure and produce all the results.

IV. Full-Wave and Experimental Results

The geodesic H-plane horn array prototype manufactured using LPBF-AM is shown in Fig. 7(a). The antenna geometries and

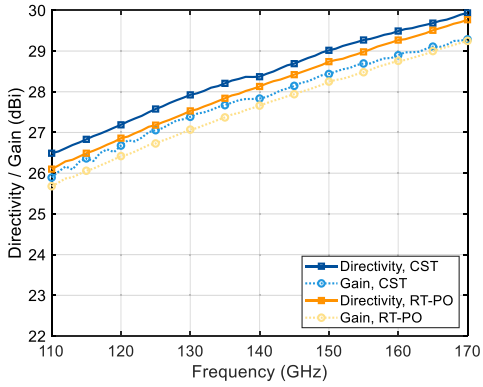


Fig. 6. Directivity and gain versus frequency of the final design obtained using CST and RT-PO.

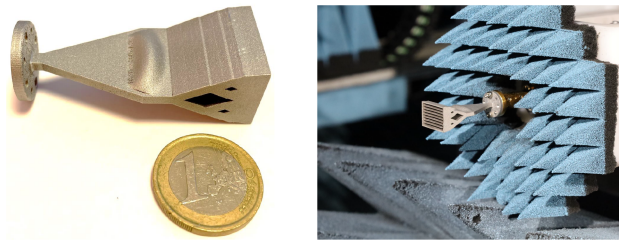


Fig. 7. Photographs of the (a) prototype and (b) measurement setup in the anechoic chamber.

TABLE I
Measured Width and Height of the Feed Waveguide at Three Different Positions Using a Light Microscope

	Mea. value (μm)	Error (μm)	Ave. error (μm)
Width	1660	10	13
	1660	10	
	1670	20	
Height	860	30	30
	870	40	
	850	20	

their selection criteria are explained in Section III. The building material is AlSi10Mg aluminum alloy, which has remarkable electromagnetic and mechanical properties in addition to being very lightweight [25], [43], [44]. The specifics of the LPBF-AM process are described in [25]. It is important to emphasize that this manufacturing technique achieves a printing precision of $\pm 50\,\mu$m and results in a surface roughness of $3\,\mu$m after post-processing. Furthermore, the designed component is meant to be manufactured monolithically, ensuring that no misalignment or leakage issues are present, specially at high frequencies, such as the *D*-band. Moreover, the compactness of the prototype can be appreciated compared to a 1€ coin. In order to assess the prototype's printing precision, a light microscope is employed to gauge the feed waveguide's width and height, with the results presented in Table I. We compare at three different positions the measured (Mea.) values with the designed ones. The average (Ave.) width and height errors are 13 and 30 μm, respectively, showing good printing precision and tolerances

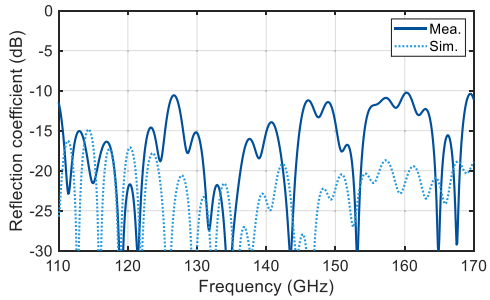


Fig. 8. Measured and simulated reflection coefficients of the prototype.

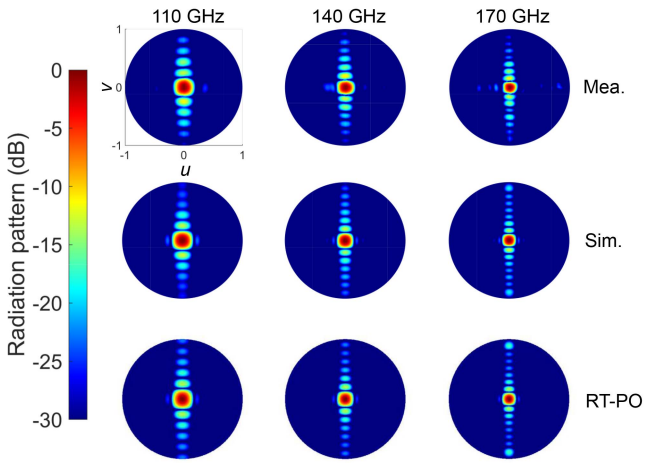


Fig. 9. Contour maps of the radiation patterns in the uv-plane at 110, 140, and 170 GHz obtained using measurement, CST simulation, and the RT-PO model.

meeting $\pm 50\,\mu$m. The prototype measurements (including the reflection coefficient, the radiation patterns, and the realized gains) were carried out in the antenna measurement facilities of the Smart Wireless Technologies Lab at the University of Granada [45], as shown in Fig. 7(b).

The measured and simulated reflection coefficients of the prototype in the operating frequency range of 110 – 170 GHz are plotted in Fig. 8. The simulated result is smaller than -15 dB in the entire *D*-band. In comparison, the measured result is better than -10 dB. The deterioration in measurement is attributed to small printing deviations. These minor manufacturing imperfections impact the antenna at high frequencies due to its small dimensions.

The contour maps of the radiation patterns in the uv-plane at 110, 140, and 170 GHz are presented in Fig. 9. The first, second, and third rows correspond to the measured, simulated, and calculated results, respectively. First, a symmetric pencil beam can be observed at all frequencies. It is worth mentioning that even though the geodesic horn shape is designed to have the same E- and H-plane beamwidths at the central frequency, the equal-beamwidth property is maintained in the whole *D*-band due to the quasioptical nature of the structure. Second, good agreement is obtained between the measured and simulated patterns with small discrepancies, which confirm the successful manufacturing of the prototype. Finally, we can appreciate the remarkable agreement between the simulated and RT-PO results

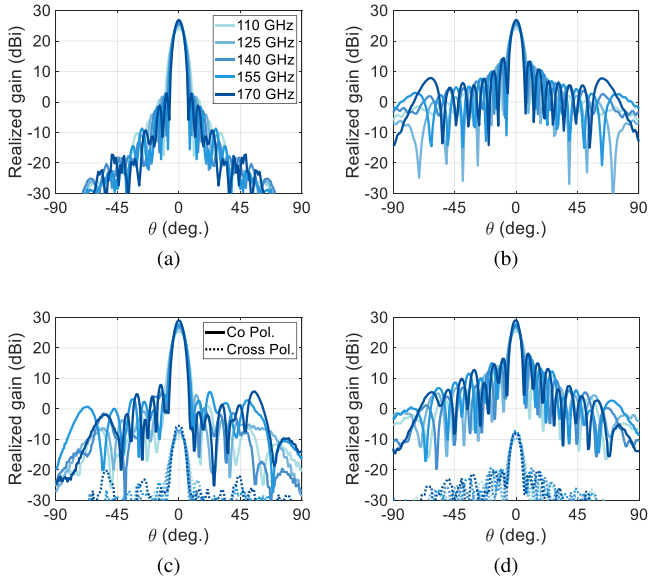


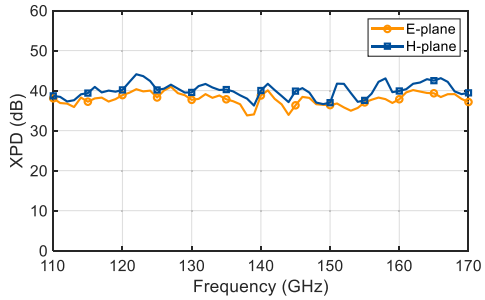
Fig. 10. Realized gain patterns over frequency in the main cuts. (a) Simulated results in the H-plane. (b) Simulated results in the E-plane. (c) Measured results in the H-plane. (d) Measured results in the E-plane. Co- and cross-polarized patterns are shown with solid and dotted lines, respectively.

in the complete uv-plane, once again demonstrating the accuracy of the proposed approximate approach.

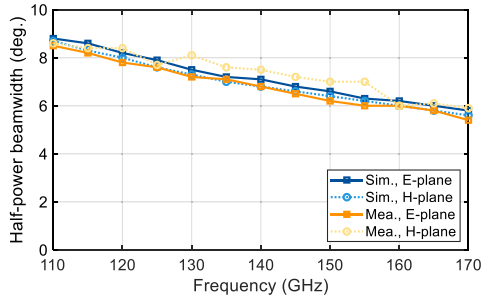
In Fig. 10 it is shown the measured and simulated realized gain patterns at different frequencies in the H- and E-planes. In the H-plane, the measured co-polarized main beams align well with the simulated counterparts, although some irregular side lobes occur in the measured data. However, all irregular lobes are at least 20 dB lower than the maximum gain values. These irregularities are primarily due to manufacturing inaccuracies and can be reduced through high-precision fabrication methods, such as microlaser sintering (MLS) [46], [47], [48], [49], [50]. Moreover, similar alignment can be observed in the E-plane results with small discrepancies in the side lobes. The simulated cross-polarized results are extremely low compared to the co-polarized components and thus not displayed in the figures. The cross-polarized components measured are shown in Fig. 10(c) and (d) with dotted lines, which are at least 30 dB lower than the maximum co-polarized gain values, demonstrating a high level of cross-polarization discrimination (XPD) of the design.

Next, Fig. 11(a) and (b) present the XPD levels and 3 dB beamwidths of the design over frequency, respectively. The measured XPD levels in both E- and H-planes are better than 33 dB at all operating frequencies. The 3 dB beamwidths along the two principal cuts are typically equal, exhibiting a maximum variance of 0.4° in simulations and 1° in experimental measurements. The larger disparity observed in the measurements is attributed to manufacturing inaccuracies and unavoidable measurement errors, leading to a slight expansion of the beamwidths in the H-plane. Both the XPD and beamwidth results exhibit consistent stability across the operating frequency bandwidth.

The detailed frequency behavior of the maximum realized gain and directivity for the horn array design is presented in Fig. 12(a). In the CST simulation, the lossy aluminum alloy of

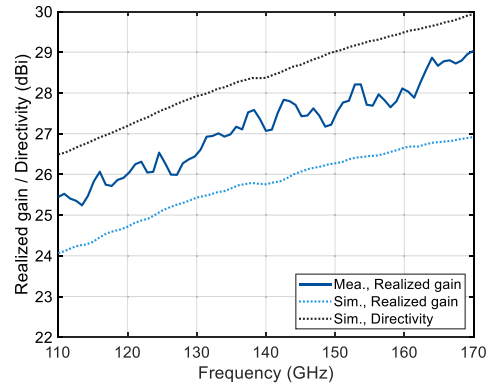


(a)

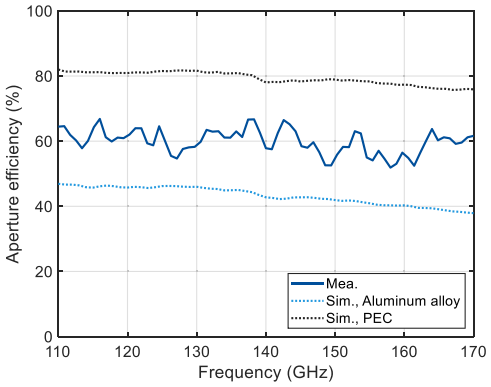


(b)

Fig. 11. (a) Measured XPD level over frequency at the broadside direction in E- and H-planes. (b) Measured and simulated 3 dB beamwidth over frequency in E- and H-planes.



(a)



(b)

Fig. 12. Measured and simulated (a) realized gain / directivity and (b) aperture efficiency versus frequency of the prototype.

TABLE II
COMPARISON WITH SOME RELEVANT HORN DESIGNS

Ref.	[19]	[51]	[34](5)	**This work**
Topology	Horn with conformal metal lens	Pyramidal horn WR-06	Diagonal horn WR-0.65	Geodesic horn array
Manuf.	Milling	Milling	Milling	LPBF
Freq. (GHz)	14.3 – 60	110 – 170	1100 – 1700	110 – 170
Gain (dBi)	12.1 – 21.7	24.8 – 26.8(2)	24.2 – 27.3	25.2 – 29
Apert. eff.(1)	46%	45%(3)	$\approx 100\%$	60%
XPD (dB)	> 25	> 60(4)	≈ 10	> 33
Axial length (λ_0)	9.7	24.2	51.3	25.7
Apert. size (λ_0^2)	6.6×1.2	9.9×7.1	5.1×5.1	9.9×7.2
Sym. beam	No	No	Yes	Yes

(1): evaluated using measured realized gain at the central frequency.
(2): simulated directivity at the central frequency.
(3): calculated using simulated directivity.
(4): simulated.
(5): all antenna parameters are based on theoretical analysis.

$\sigma = 1.68 \times 10^7$ S/m with an assumed surface roughness of 3 μm is used as the building material. Since it is well understood that the surface roughness model in CST is only valid for normally distributed surfaces at low frequencies [52], the simulated results are only provided as a reference and are not expected to reflect the actual surface roughness of the prototype. The measured realized gain falls between the simulated directivity and the realized gain with some variations caused by random errors in the measurement. Within an uncertainty margin of ± 0.3 dB, the measured realized gain increases from 25.5 dBi at 110 GHz to 29 dBi at 170 GHz with an estimated average insertion loss of 1.23 dB resulting from the finite conductivity and surface roughness of the printing material. We can infer from the measured results that the insertion loss of the horn array is overestimated in the CST simulation at such high frequencies. In fact, the estimated radiation efficiency in the measurement ranges from 67% to 86%. The corresponding aperture efficiency results are presented in Fig. 12(b). The measured aperture efficiency calculated using realized gain reaches on average 60% throughout the *D*-band.

The key parameters of our design are compared with those of some other relevant horn designs in Table II. In [19], by integrating a conformal metal lens into a H-plane horn, the design realizes high gains in a very broad bandwidth. In comparison, our proposed antenna has a higher gain and a symmetric beam due to the inclusion of the array in the E-plane. Another relevant horn design is the pyramidal horn, a simple and well-established structure, such as the commercial example provided in [51]. This pyramidal horn has the same operating frequency as our design, with a similar axial length and aperture size. However, our design has higher gain and aperture efficiency due to correction of phase errors. Note that the aperture efficiency of 45% of the pyramidal horn is calculated using simulated directivity, while our design has an aperture efficiency of nearly 80% with simulated directivity. The commercial diagonal horn that operates above 1 THz in [34] has high gain, a high aperture efficiency, and a symmetric beam. However, the diagonal horn has twice the axial length of our design and exhibits inferior XPD performance. Therefore, our proposed antenna represents a compact solution with well-balanced performance and low-cost realization.

TABLE III
COMPARISON WITH SOME OTHER HIGH-GAIN SUB-THZ ANTENNAS

Reference	Topology	Fabrication	Freq. (GHz)	Max. gain (dBi)	[1]Max. aper. eff.	Max. rad. eff.	XPD (dB)	Sym. beam
[7]	SIW feed array	LTCC	$150-170$	16.8	N/A	55%	>30	No
[8]	Pillbox BFN feed [2]LWA	MEMS micromachining	$220-300$	28.5	45%	$\approx 84\%$	>35	No
[9]	Air-filled plate array	MEMS micromachining	$120-160$	33.2	$\approx 70\%$	71%	>38	Yes
[10]	Corporate feed array	Diffusion bonding	$131-150$	26.4	67%	N/A	>45	Yes
[11]	Gap waveguide array	CNC machining	$124-152$	31.5	94%	94%	>33	Yes
[13]	Dielectric lens	CNC micromachining	$230-310$	30	N/A	$\approx 60\%$	N/A	No
This work	Geodesic horn array	LPBF-AM	$110-170$	29	67%	86%	>33	Yes

[1]: calculated using realized gain.
[2]: leaky-wave antenna.

Finally, we summarize in Table III the main performance of the proposed geodesic H-plane horn array antenna in comparison with some other high-gain sub-THz antennas. Our design achieves high gain, symmetric beam, high efficiencies, and high XPD level in a wide frequency range (43% fractional bandwidth). The metal-only design possesses superior mechanical and thermal properties, which ensures a good environmental stability and makes it highly suitable for space and high-power applications [53]. Owing to its quasioptical characteristics, the proposed antenna is susceptible of being scaled in frequency with ease. Nevertheless, the primary constraint we found arises from the LPBF-AM process employed, particularly concerning finite conductivity of the printing material ($\sigma = 1.68 \times 10^7$ S/m), the achieved surface roughness ($3\,\mu$m), and its printing accuracy ($\pm 50\,\mu$m). The gold coating technique can be used in postprocessing to increase the conductivity of the inner surface [49], while electrochemical polishing is a viable solution to further reduce its surface roughness [54]. Moreover, an enhanced iteration of LPBF-AM termed the high-precision MLS technique, or microselective laser melting, is capable of reaching dimensional precision within $\pm 15\,\mu$m and attaining a surface roughness below $1\,\mu$m [46]. The application of the MLS technique to microwave components has been shown to produce waveguide devices functioning at frequencies up to approximately 300 GHz, while maintaining minimal return and insertion losses [47], [48]. Consequently, by employing MLS in its fabrication and applying the corresponding postprocessing procedures, the proposed antenna has the potential to be extended to a comparable frequency range. The estimates of Northern Waves AB confirm that this technology ensures the practical and scalable production of advanced sub-THz antennas, establishing LPBF-AM as a cornerstone for the next generation of high-performance wireless systems [55].

V. CONCLUSION

In this work, we present the design of a high-gain symmetric-beam geodesic H-plane horn antenna operating in the *D*-band. An RT-PO model is elaborated to facilitate the design, which can calculate the full radiation pattern, directivity, gain, and conductive loss of any PPW-based array antenna. The design is specially adapted to be manufactured monolithically using the LPBF-AM technique. The experimental results show excellent performance of the prototype up to 170 GHz, which marks the first successful application of the LPBF-AM technique to a geodesic design at such high frequencies. This article showcases the great potential of LPBF-AM for fast and low-cost prototyping of sub-THz metallic antennas and other microwave components. The proposed low-cost high-gain fully metallic antenna is well suited to establish stable wireless connections in harsh environments. Potential applications of the proposed antenna include high-power radar systems, high-resolution imaging systems, and high-speed satellite and terrestrial wireless communication systems at sub-THz. Future work includes moving to even higher frequencies and increasing the aperture size to achieve higher gain.

REFERENCES

[1] K. B. Cooper et al., "THz imaging radar for standoff personnel screening," *IEEE Trans. Terahertz Sci. Technol.*, vol. 1, no. 1, pp. 169–182, Sep. 2011.
[2] J. Grzyb, K. Statnikov, N. Sarmah, B. Heinemann, and U. R. Pfeiffer, "A 210– 270-GHz circularly polarized FMCW radar with a single-lens-coupled SiGe HBT chip," *IEEE Trans. Terahertz Sci. Technol.*, vol. 6, no. 6, pp. 771–783, Nov. 2016.
[3] T. S. Rappaport et al., "Wireless communications and applications above 100 GHz: Opportunities and challenges for 6G and beyond," *IEEE Access*, vol. 7, pp. 78729–78757, Jun. 2019.
[4] A. Hirata et al., "120-GHz-band wireless link technologies for outdoor 10-Gbit/s data transmission," *IEEE Trans. Microw. Theory Techn.*, vol. 60, no. 3, pp. 881–895, Mar. 2012.
[5] C. De Lima et al., "Convergent communication, sensing and localization in 6G systems: An overview of technologies, opportunities and challenges," *IEEE Access*, vol. 9, pp. 26902–26925, Feb. 2021.
[6] J. Xu, Z. N. Chen, X. Qing, and W. Hong, "140-GHz planar broadband LTCC SIW slot antenna array," *IEEE Trans. Antennas Propag.*, vol. 60, no. 6, pp. 3025–3028, Jun. 2012.
[7] S. Zhao, Y. Xu, and Y. Dong, "An LTCC-based antenna array for D-band terahertz communication," *IEEE Antennas Wireless Propag. Lett.*, vol. 22, no. 6, pp. 1491–1495, Jun. 2023.
[8] A. Gomez-Torrent et al., "A low-profile and high-gain frequency beam steering subterahertz antenna enabled by silicon micromachining," *IEEE Trans. Antennas Propag.*, vol. 68, no. 2, pp. 672–682, Feb. 2020.
[9] S. S. Yao, Y. J. Cheng, M. M. Zhou, Y. F. Wu, and Y. Fan, "D-band wideband air-filled plate array antenna with multistage impedance matching based on MEMS micromachining technology," *IEEE Trans. Antennas Propag.*, vol. 68, no. 6, pp. 4502–4511, Jun. 2020.
[10] Y. J. Cheng, F. Y. Tan, M. M. Zhou, and Y. Fan, "Dual-polarized wideband plate array antenna with high polarization isolation and low cross polarization for D-band high-capacity wireless application," *IEEE Antennas Wireless Propag. Lett.*, vol. 19, no. 12, pp. 2023–2027, Dec. 2020.
[11] A. Farahbakhsh, D. Zarifi, and A. Uz Zaman, "D-band high-gain planer slot array antenna using gap waveguide technology," *IEEE Trans. Antennas Propag.*, vol. 73, no. 1, pp. 594–599, Jan. 2025.
[12] A. Garcia-Pino, N. Llombart, B. Gonzalez-Valdes, and O. Rubinos-Lopez, "A bifocal ellipsoidal Gregorian reflector system for THz imaging applications," *IEEE Trans. Antennas Propag.*, vol. 60, no. 9, pp. 4119–4129, Sep. 2012.

[13] K. Konstantinidis et al., "Low-THz dielectric lens antenna with integrated waveguide feed," *IEEE Trans. Terahertz Sci. Technol.*, vol. 7, no. 5, pp. 572–581, Sep. 2017.
[14] N. v. Rooijen et al., "A core-shell lens for antenna on-package integration at D-band," *IEEE Trans. Antennas Propag.*, vol. 72, no. 8, pp. 6195–6208, Aug. 2024.
[15] S. Bosma et al., "Dual-band submillimeter-wave leaky-wave lens antenna for heterodyne cometary mapping," *IEEE Trans. Terahertz Sci. Technol.*, vol. 15, no. 2, pp. 269–282, Mar. 2025.
[16] Y. J. Guo, M. Ansari, R. W. Ziolkowski, and N. J. G. Fonseca, "Quasi-optical multi-beam antenna technologies for B5G and 6G mmWave and THz networks: A review," *IEEE Open J. Antennas Propag.*, vol. 2, pp. 807–830, Jul. 2021.
[17] O. Quevedo-Teruel et al., "Geodesic lens antennas for 5G and beyond," *IEEE Commun. Mag.*, vol. 60, no. 1, pp. 40–45, Jan. 2022.
[18] M. Chen, F. Mesa, and O. Quevedo-Teruel, "Geodesic H-plane horn antennas," *IEEE Trans. Antennas Propag.*, vol. 71, no. 8, pp. 6329–6339, Aug. 2023.
[19] Z.-Y. Zhang and K. W. Leung, "Wideband horn antenna with integrated conformal metal lens," *IEEE Trans. Antennas Propag.*, vol. 72, no. 7, pp. 5651–5662, Jul. 2024.
[20] P. Castillo-Tapia et al., "SubTHz fully-metallic geodesic Luneburg lens antenna," *IEEE Trans. Terahertz Sci. Technol.*, vol. 15, no. 3, pp. 514–518, May 2025.
[21] R. Rinehart, "A solution of the problem of rapid scanning for radar antennae," *J. Appl. Phys.*, vol. 19, no. 9, pp. 860–862, Sep. 1948.
[22] K. S. Kunz, "Propagation of microwaves between a parallel pair of doubly curved conducting surfaces," *J. Appl. Phys.*, vol. 25, no. 5, pp. 642–653, May 1954.
[23] J. Rico-Fernández, F. V. Vidarsson, M. Arrebola, N. J. G. Fonseca, and O. Quevedo-Teruel, "Compact and lightweight additive manufactured parallel-plate waveguide half-Luneburg geodesic lens multiple-beam antenna in the Ka-band," *IEEE Antennas Wireless Propag. Lett.*, vol. 22, no. 4, pp. 684–688, Apr. 2023.
[24] P. Castillo-Tapia, J. Rico-Fernández, S. Clendinning, F. Mesa, and O. Quevedo-Teruel, "Evaluation of losses in 3-D-printed geodesic lenses using a ray-tracing model," *IEEE Trans. Antennas Propag.*, vol. 72, no. 1, pp. 234–242, Jan. 2024.
[25] M. Chen, J. Rico-Fernández, F. Mesa, and O. Quevedo-Teruel, "Metal-only additive manufacturing of V-band lightweight waveguide and horn components," *IEEE Trans. Microw. Theory Techn.*, vol. 73, no. 9, pp. 5675–5685, Sep. 2025.
[26] S. Clendinning, O. Zetterstrom, J. Rico-Fernández, F. Mesa, and O. Quevedo-Teruel, "Nonrotationally symmetric V-band geodesic lens antenna with defined footprint," *IEEE Trans. Antennas Propag.*, vol. 73, no. 4, pp. 2648–2653, Apr. 2025.
[27] M. Chen, E. Miquel-Nardi, J. Rico-Fernández, C. Segura-Gómez, and O. Quevedo-Teruel, "W-band compact and lightweight additively manufactured geodesic H-plane horn antenna," *IEEE Antennas Wireless Propag. Lett.*, early access, pp. 1–5, doi: 10.1109/LAWP.2025.3561471.
[28] P. Castillo-Tapia et al., "Two-dimensional beam steering using a stacked modulated geodesic Luneburg lens array antenna for 5G and beyond," *IEEE Trans. Antennas Propag.*, vol. 71, no. 1, pp. 487–496, Jan. 2023.
[29] P. Castillo-Tapia, J. Rico-Fernández, and O. Quevedo-Teruel, "V-band monolithic additive-manufactured geodesic lens array antenna," *IEEE Antennas Wireless Propag. Lett.*, vol. 22, no. 10, pp. 2527–2531, Oct. 2023.
[30] Y. E. Yamac, A. Caliskan, A. S. Turk, and A. Kizilay, "Waveguide slot-fed horn antenna array with suppressed sidelobes," in *Proc. 2019 Signal Process. Symp.*, pp. 7–10, Oct. 2019.
[31] S. Withington and J. Murphy, "Analysis of diagonal horns through Gaussian-Hermite modes," *IEEE Trans. Antennas Propag.*, vol. 40, no. 2, pp. 198–206, Feb. 1992.
[32] J. Johansson and N. Whyborn, "The diagonal horn as a sub-millimeter wave antenna," *IEEE Trans. Microw. Theory Techn.*, vol. 40, no. 5, pp. 795–800, May 1992.
[33] H. J. Gibson et al., "A novel spline-profile diagonal horn suitable for integration into THz split-block components," *IEEE Trans. Terahertz Sci. Technol.*, vol. 7, no. 6, pp. 657–663, Nov. 2017.
[34] Virginia Diodes, Inc., "Nominal horn specifications," Charlottesville, Virginia, USA. [Online]. Available: https://www.vadiodes.com/images/AppNotes/VDI_Feedhorn_Summary_2020.05.04.pdf
[35] Q. Liao et al., "Ray-tracing model for generalized geodesic-lens multiple-beam antennas," *IEEE Trans. Antennas Propag.*, vol. 71, no. 3, pp. 2640–2651, Mar. 2023.
[36] D. Kirsanov, G. Malhorta, and S. Knock, "gdist." [Online]. Available: https://github.com/the-virtual-brain/tvb-gdist
[37] S. A. Schelkunoff, "Some equivalence theorems of electromagnetics and their application to radiation problems," *Bell Syst. Tech. J.*, vol. 15, no. 1, pp. 92–112, Jan. 1936.
[38] C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed. Hoboken, NY, USA: Wiley, 2016.
[39] M. Chen, O. Habiboglu, F. Mesa, and O. Quevedo-Teruel, "Ray-tracing and physical-optics model for planar Mikaelian lens antennas," *IEEE Trans. Antennas Propag.*, vol. 72, no. 2, pp. 1735–1744, Feb. 2024.
[40] S. Orfanidis, Electromagnetic waves and antennas, 2016, pp. 544–552. [Online]. Available: https://www.ece.rutgers.edu/orfanidi/ewa/
[41] F. Mesa, M. Chen, P. Castillo-Tapia, and O. Quevedo-Teruel, "Physical optics applied to parallel-plate lens antennas," *IEEE Open J. Antennas Propag.*, vol. 5, no. 4, pp. 833–844, Aug. 2024.
[42] D. M. Pozar, *Microwave Engineering: Theory and Techniques*. Hoboken, NJ, USA: John Wiley & Sons, Inc., 2021.
[43] C. Silbernagel, I. Ashcroft, P. Dickens, and M. Galea, "Electrical resistivity of additively manufactured AlSi10Mg for use in electric motors," *Additive Manuf.*, vol. 21, pp. 395–403, May 2018.
[44] G. Addamo, L. Scalcinati, M. Zannoni, O. A. Peverini, and F. Calignano, "Feasibility study on additive manufacturing of feed horn operating in D-band," *Sensors*, vol. 25, no. 2, Jan. 2025, Art. no. 523.
[45] Smart Wireless Technologies Lab–Singular Laboratories. [Online]. Available: https://investiga.ugr.es/laboratorio-singular/smart-wireless-technologies-lab/
[46] B. Nagarajan, Z. Hu, X. Song, W. Zhai, and J. Wei, "Development of micro selective laser melting: The state of the art and future perspectives," *Engineering*, vol. 5, no. 4, pp. 702–720, Aug. 2019.
[47] V. Fiorese et al., "Evaluation of micro laser sintering metal 3D-printing technology for the development of waveguide passive devices up to 325 GHz," in *Proc. 2020 IEEE/MTT-S Int. Microw. Symp.*, Oct. 2020, pp. 1168–1171.
[48] T. Skaik et al., "A 3-D printed 300 GHz waveguide cavity filter by micro laser sintering," *IEEE Trans. Terahertz Sci. Technol.*, vol. 12, no. 3, pp. 274–281, May 2022.
[49] T. Skaik et al., "Evaluation of 3-D printed monolithic G-band waveguide components," *IEEE Trans. Compon Packag. Manuf. Technol.*, vol. 13, no. 2, pp. 240–248, Feb. 2023.
[50] L. Qian, T. Skaik, Y.-W. Wu, and Y. Wang, "Three-dimensional printed mm-wave monolithic waveguide twists: End-to-end manufacturing and surface-treatment solutions," *IEEE Trans. Microw. Theory Techn.*, vol. 73, no. 9, pp. 5909–5917, Sep. 2025, doi: 10.1109/TMTT.2025.3577733.
[51] Anteral, "D-band standard gain horn." [Online]. Available: https://anteral.com/datasheets/standard-gain-horn-antenna-wr06-26-dbi-gain.pdf
[52] A. Hofmann et al., "SLS-printed E-band waveguides and the impact of surface roughness," in *Proc. 2023 53 rd Eur. Microw. Conf.*, Sep. 2023, pp. 243–246.
[53] H. Talpur, U. Myrzakhan, J. A. Vásquez-Peralvo, S. Zhang, and S. Chatzinotas, "Additive manufacturing of antennas and RF components for SATCOM: A review," *IEEE Open J. Antennas Propag.*, vol. 6, no. 4, pp. 943–977, Aug. 2025.
[54] U. Ali, H. Fayazfar, F. Ahmed, and E. Toyserkani, "Internal surface roughness enhancement of parts made by laser powder-bed fusion additive manufacturing," *Vacuum*, vol. 177, Jul. 2020, Art. no. 109314.
[55] Northern waves AB. [Online] Available: https://northern-waves.com/



Mingzheng Chen (Member, IEEE) received the Ph.D. degree in electromagnetic engineering from the KTH Royal Institute of Technology, Stockholm, Sweden, in 2025.

He is currently a Postdoctoral Researcher with the KTH Royal Institute of Technology. He has authored or coauthored more than 50 peer-reviewed journal and conference papers. His research interests include quasioptical structures, periodic structures, and additive manufacturing.

Dr. Chen was the recipient of the 2023 Best Paper Award from *National Science Review* and the Best Paper Award at MTTW 2023 for his works.



José Rico-Fernández (Member, IEEE) was born in Oviedo, Spain, in 1997. He received the B.Sc., M.Sc., and Ph.D. degrees in telecommunications engineering from the Universidad de Oviedo, Gijón, Spain, in 2019, 2021, and 2024 respectively.

Since 2023, he has been a Chief Executive Officer with Northern Waves AB, Stockholm, Sweden. His research interests include the design and production of additive-manufactured microwave components and antennas in mm-wave frequencies.



Hairu Wang (Graduate Student Member, IEEE) received the bachelor's degrees in information engineering from the Southeast University, Nanjing, China, in 2021, and the master 's degrees in science from the KTH Royal Institute of Technology, Stockholm, Sweden, in 2022. She is currently working toward the Ph.D. degree in antennas and electromagnetics with the Division of Electromagnetic Engineering and Fusion Science, KTH.

Her research interests include periodic structures and ray-tracing and physical-optics modeling.



Cleofás Segura-Gómez (Graduate Student Member, IEEE) was born in Almería, Spain, in 1997. He received the B.Sc. and M.Sc. degrees in telecommunication engineering, and the Ph.D. degree in information and communication technologies from the University of Granada (UGR), Granada, Spain, in 2019, 2021, and 2025, respectively.

From 2019 to 2025, he held several predoctoral research contracts and substitute teaching positions with the Department of Signal Theory, Telematics, and Communications. In 2023, he conducted a predoctoral research stay with Heriot-Watt University, Edinburgh, U.K. His research interests include millimeter-wave components (filters, antennas, and phase shifters), antenna arrays, leaky-wave antennas, and reconfigurable planar devices.

Dr. Segura-Gómez has been a Member of the SWAT-UGR Group, focusing on the design, fabrication, and characterization of RF devices at very high frequencies, since 2019.



Francisco Mesa (Fellow, IEEE) received the Licenciado and Ph.D. degrees in physics from the Universidad de Sevilla, Seville, Spain, in 1989 and 1991, respectively.

He is currently a Professor with the Departamento de Física Aplicada 1, Universidad de Sevilla. He is also Affiliate Professor with the KTH Royal Institute of Technology, Stockholm, Sweden. His research interests include electromagnetic propagation/radiation in microwave and quasioptical structures.



Oscar Quevedo-Teruel (Fellow, IEEE) received the Telecommunication Engineering and Ph.D. degrees from the Carlos III University of Madrid, Madrid, Spain, in 2005 and 2010, respectively.

From 2010 to 2011, he was a Research Fellow with the Department of Theoretical Physics of Condensed Matter, Universidad Autonoma de Madrid, Madrid, Spain, and was a Postdoctoral Research with the Queen Mary University of London, London, U.K., from 2011 to 2013. In 2014, he was with the KTH Royal Institute of Technology, Stockholm, Sweden, where he is currently a Professor with the Division of Electromagnetic Engineering and Fusion Science and the Director of the master's programme in electromagnetics fusion and space engineering. He has coauthored more than 150 papers in international journals and 250 papers at international conferences. He has made scientific contributions to higher symmetries, transformation optics, ray tracing, lens antennas, metasurfaces, and high-impedance surfaces.

Dr. Quevedo-Teruel has been a Member of the European Association on Antennas and Propagation (EurAAP) Board of Directors since January 2021. Since January 2022, he has been the Vice-Chair of EurAAP. He was a Distinguished Lecturer of the IEEE Antennas and Propagation Society, from 2019 to 2021. He has been an Associate Editor for IEEE TRANSACTIONS ON ANTENNAS AND PROPAGATION, from 2018 to 2022, and a Track Editor since 2022.